

Dhrubojeet Haldar

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Summary

Financial Mathematics graduate student with experience in systematic long/short equity strategies, equity futures microstructure research and inventory-aware market-making (C++20). Background in quantitative model validation with emphasis on regime-conditional signal diagnostics.

Education

North Carolina State University	Raleigh, NC
Master of Financial Mathematics	12/2026
GPA: 3.82 /4.0	
Indian Institute of Technology, Roorkee	Roorkee, India
Bachelor of Technology in Computer Science	07/2023

Related Coursework

Stochastic Calculus for Finance | Options and Derivatives Pricing | Monte Carlo Methods for Finance
Probability and Stochastics for Finance | Linear Models and Regression | Statistical Inference
Data Structures | Statistical Learning

Skills

Programming: Python, C++20, SAS, SQL, R
Packages: Pandas, SciPy, NumPy, Scikit-learn, Statsmodels, NLOpt, VolLib, QuantLib, XGBoost
Quantitative Methods: Time Series Analysis, Monte Carlo Simulation, Numerical Optimization
Markets and Data: Bloomberg Terminal, yfinance, Databento (MBP-10), QuantConnect

Experience

Quantitative Model Solutions Specialist Wells Fargo	Bangalore, India 06/2023 – 07/2025
- Diagnosed regime-conditional signal decay in Cards (\$48B) credit models using KS and PSI drift metrics. - Identified feature-level drift predicting scoring model instability ahead of formal model review cycles. - Quantified model sensitivity to macro and behavioral regime shifts across multi-million-record datasets.	

Project Experience

Avellaneda-Stoikov Market-Making - BTCUSD Perpetual Futures North Carolina State University, Financial Mathematics Department	06/2026 – Present
- Built AS market-maker (C++20) on BTCUSD perpetual; MLE-calibrated fill intensity; 105 walk-forward windows. - Real-tape fill replay (not Poisson); order latency and glitch filter; hard inventory bounds (± 5 lots). Vol-adaptive horizon and tight inventory bounds: reduced losses by 58%, max drawdown by 47%, avg realized spread of -\$63.51 against -\$352.05 of fixed-horizon baseline. - Net PnL -\$73.95 (936 fills) pre-rebate: residual loss traced to low-vol regime (W37) and absent queue model.	
Russell 2000 Deep Momentum - XGBoost 4-Quadrant Long/Short Reclassifier North Carolina State University, Financial Mathematics Department	05/2026 – 06/2026
- Bimodality-motivated 4-quadrant XGBoost reclassifier on 12-1 momentum (Russell 2000, \$10M, monthly rebalance); 10-member ensemble, 36-month rolling window, 19 cross-sectionally ranked features. - Walk-forward (147 rebalances): Sharpe 0.235 vs 0.126 baseline; drawdown -8.1pp; vol reduced by 40%. - Primary alpha: GL quadrant (bottom-decile reversal, +1.90%/mo) – orthogonal to momentum factor. - Diagnosed post-2020 Sharpe collapse (+0.418 to -0.042) : rate-hike beta drag, not signal decay; L/S spread intact (+0.85%/mo post vs +0.86%/mo pre-2020); vol_12m importance (0.071 to 0.116), displacing cross-sectional signals.	
ES Futures Calendar Spread Mean-Reversion North Carolina State University, Financial Mathematics Department	03/2026 – 05/2026
- ES calendar spread mean-reversion via Fair Value deviation (SOFR + div yield + expiry differential); Databento MBP-10 nanosecond data, 4 quarterly roll windows (Sep 24-Jun 25). - High-conviction entry at $z_fill > 3$ (bias-corrected); avg net +\$2.86/lot net of costs. - European session is a significant alpha source (n=170, p = 0.0008, avg +\$4.9/trade) across all 4 windows. - OOS results: p = 0.0265 (n=670); drift-4h gate halved σ per trade; per-trade mean annualized Sharpe +3.88 (CI [+1.38, +6.63]), net of \$8.04/lot (exchange, NFA and broker fees).	
IMC Trading Prosperity 4 - Algorithmic Trading Competition	03/2026
Ranked 42nd globally in Phase 1 (~18k teams); achieved final rank of ~900 (top 5%) across all participants.	